

HERO scorecard (vertical) — value ± BCa 95% CI · colour = per-metric min-max (green = best) · * = Δ-vs-frozen CI excludes 0
13 metrics × 14 encoders · WINNER col + BLUE BOX = surgery-vs-vanilla-cont-SSL champion duel (tie = paired 95% CI overlaps 0) · 10 K bootstrap

	g-frozen	g-fullFT	g-LPFT	g-DoRA	g-LoRA	g-vCSSL-enc	g-raw	g-surgical	g-s3DI-enc	g-surgical	g-argn	g-surgical	g-surgical	g-surgical	WINNER
action_top1 ↑ better	42.466 ±2.274	50.904 ±2.301*	46.082 ±2.321*	47.562 ±2.274*	46.685 ±2.247*	49.315 ±2.301*	48.000 ±2.274*	48.274 ±2.329*	47.671 ±2.274*	47.288 ±2.247*	49.699 ±2.274*	45.808 ±2.301*	45.096 ±2.247*	45.096 ±2.274*	tie
motion_cos ↑ better	0.007 ±0.001	0.090 ±0.003*	0.038 ±0.001*	0.068 ±0.002*	0.068 ±0.002*	0.054 ±0.002*	0.060 ±0.002*	0.048 ±0.002*	0.047 ±0.002*	0.044 ±0.002*	0.080 ±0.003*	0.024 ±0.001*	0.016 ±0.001*	0.011 ±0.001*	ViT-g surgical autorgn encoder
taxonomy_f1 ↑ better	0.802 ±0.016	0.784 ±0.017*	0.789 ±0.017	0.784 ±0.017*	0.791 ±0.017	0.789 ±0.017	0.788 ±0.017	0.790 ±0.017	0.785 ±0.017*	0.790 ±0.017	0.781 ±0.017*	0.795 ±0.017	0.794 ±0.017	0.799 ±0.016	tie
future_mse ↓ better	0.637 ±0.001	0.628 ±0.001*	0.623 ±0.001*	0.596 ±0.001*	0.596 ±0.001*	0.607 ±0.001*	0.587 ±0.001*	0.590 ±0.001*	0.591 ±0.001*	0.591 ±0.001*	0.611 ±0.001*	0.600 ±0.001*	0.607 ±0.001*	0.617 ±0.001*	ViT-g surgery raw encoder
rollout ↓ better	0.007 ±0.000	0.006 ±0.000*	0.005 ±0.000*	0.005 ±0.000*	0.005 ±0.000*	0.006 ±0.000*	0.005 ±0.000*	0.005 ±0.000*	0.006 ±0.000*	0.006 ±0.000*	0.006 ±0.000*	0.006 ±0.000*	0.006 ±0.000*	0.007 ±0.000*	ViT-g surgical 3stage DI diheavy encoder
causal ↓ better	0.655 ±0.002	0.648 ±0.002*	0.644 ±0.002*	0.617 ±0.002*	0.617 ±0.002*	0.628 ±0.002*	0.611 ±0.002*	0.614 ±0.002*	0.615 ±0.002*	0.614 ±0.002*	0.633 ±0.002*	0.622 ±0.002*	0.629 ±0.002*	0.637 ±0.002*	ViT-g surgery raw encoder
tdist ↓ better	0.007 ±0.000	0.010 ±0.000*	0.009 ±0.000*	0.009 ±0.000*	0.009 ±0.000*	0.008 ±0.000*	0.009 ±0.000*	0.009 ±0.000*	0.009 ±0.000*	0.009 ±0.000*	0.009 ±0.000*	0.009 ±0.000*	0.008 ±0.000*	0.008 ±0.000*	ViT-g surgical intervene wiseft f70 encoder
teacher_free ↓ better	0.044 ±0.001	0.039 ±0.001*	0.038 ±0.001*	0.036 ±0.001*	0.036 ±0.001*	0.037 ±0.001*	0.035 ±0.001*	0.034 ±0.001*	0.038 ±0.001*	0.038 ±0.001*	0.039 ±0.001*	0.039 ±0.001*	0.040 ±0.001*	0.041 ±0.001*	ViT-g surgical 3stage DI diheavy encoder
maskratio ↓ better	0.065 ±0.001	0.057 ±0.001*	0.056 ±0.001*	0.053 ±0.001*	0.054 ±0.001*	0.059 ±0.001*	0.053 ±0.001*	0.052 ±0.001*	0.054 ±0.001*	0.060 ±0.001*	0.062 ±0.001*	0.062 ±0.001*	0.063 ±0.001*	0.064 ±0.001*	ViT-g surgical 3stage DI diheavy encoder
aot ↑ better	0.965 ±0.006	0.945 ±0.008*	0.956 ±0.007*	0.953 ±0.007*	0.947 ±0.008*	0.950 ±0.007*	0.953 ±0.007*	0.957 ±0.007*	0.940 ±0.008*	0.952 ±0.007*	0.946 ±0.008*	0.959 ±0.007*	0.961 ±0.007	0.964 ±0.006	ViT-g surgical intervene wiseft f70 encoder
toV ↑ better	0.923 ±0.007	0.912 ±0.007*	0.925 ±0.007	0.926 ±0.007	0.922 ±0.007	0.927 ±0.007	0.916 ±0.007*	0.925 ±0.007	0.923 ±0.007	0.922 ±0.007	0.910 ±0.007*	0.925 ±0.007	0.922 ±0.007	0.919 ±0.007	tie
pace ↑ better	0.729 ±0.013	0.704 ±0.013*	0.720 ±0.013	0.722 ±0.012	0.725 ±0.013	0.701 ±0.012*	0.702 ±0.013*	0.711 ±0.012*	0.722 ±0.012	0.724 ±0.013	0.719 ±0.012	0.729 ±0.013	0.731 ±0.013	0.724 ±0.013	ViT-g surgical intervene wiseft f50 encoder
tcc_tau ↑ better	0.334 ±0.006	0.307 ±0.005*	0.290 ±0.005*	0.286 ±0.005*	0.285 ±0.005*	0.296 ±0.005*	0.289 ±0.005*	0.287 ±0.005*	0.288 ±0.005*	0.289 ±0.005*	0.292 ±0.005*	0.303 ±0.005*	0.313 ±0.006*	0.323 ±0.006*	ViT-g surgical intervene wiseft f70 encoder